

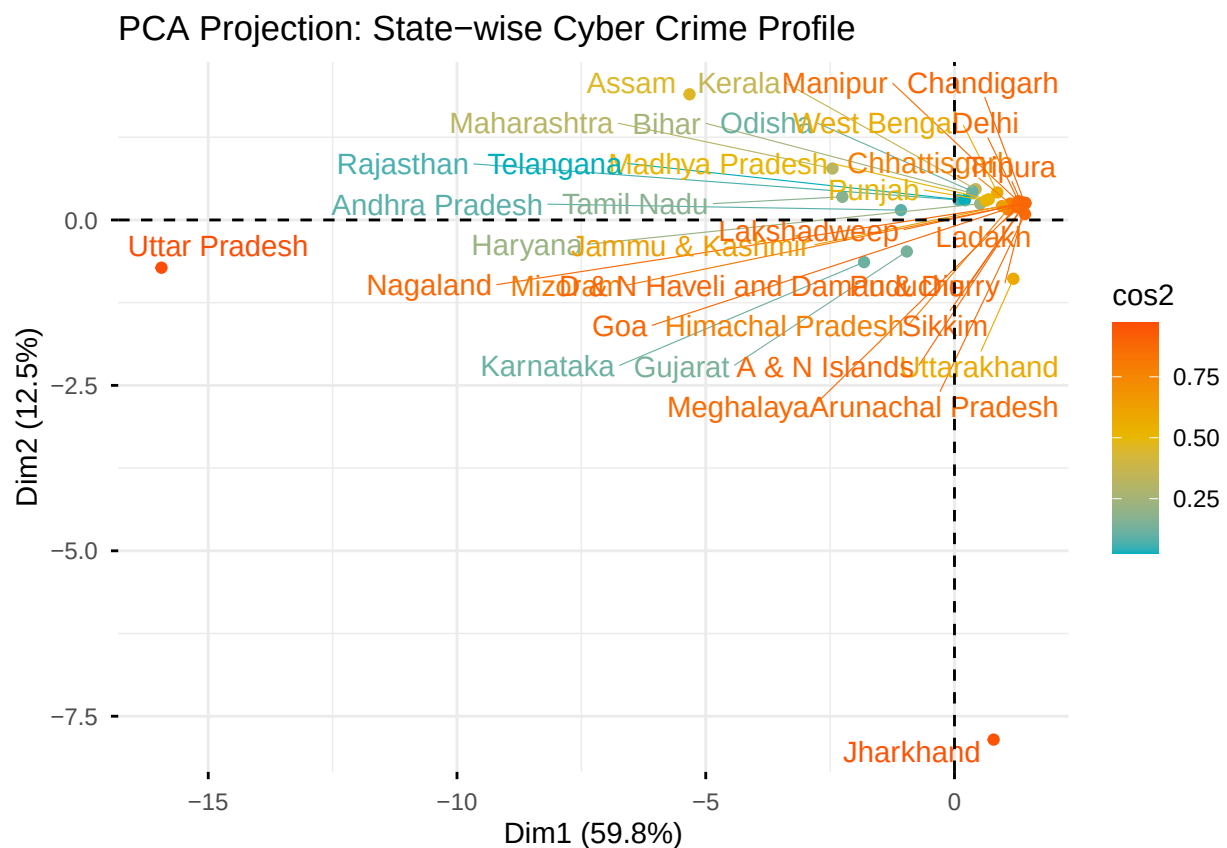
Cyber Crime Analysis Report - 2020

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(a) PCA Based Projection

```
# Load and scale data
df <- read.csv("cyber_crime_2020_1a1ee499-5095-4d7b-a2d4-412a108259da.csv")
states <- df[, 1]
data_scaled <- scale(df[, 2:ncol(df)])
rownames(data_scaled) <- states

pca_res <- prcomp(data_scaled)
fviz_pca_ind(pca_res, col.ind = "cos2",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE) +
  labs(title = "PCA Projection: State-wise Cyber Crime Profile")
```



Interpretation: The projection highlights **Uttar Pradesh** and **Jharkhand** as significant outliers. Most

other states cluster together, suggesting a common underlying structure for cyber-crime motives across India.

(b) Outlier Detection

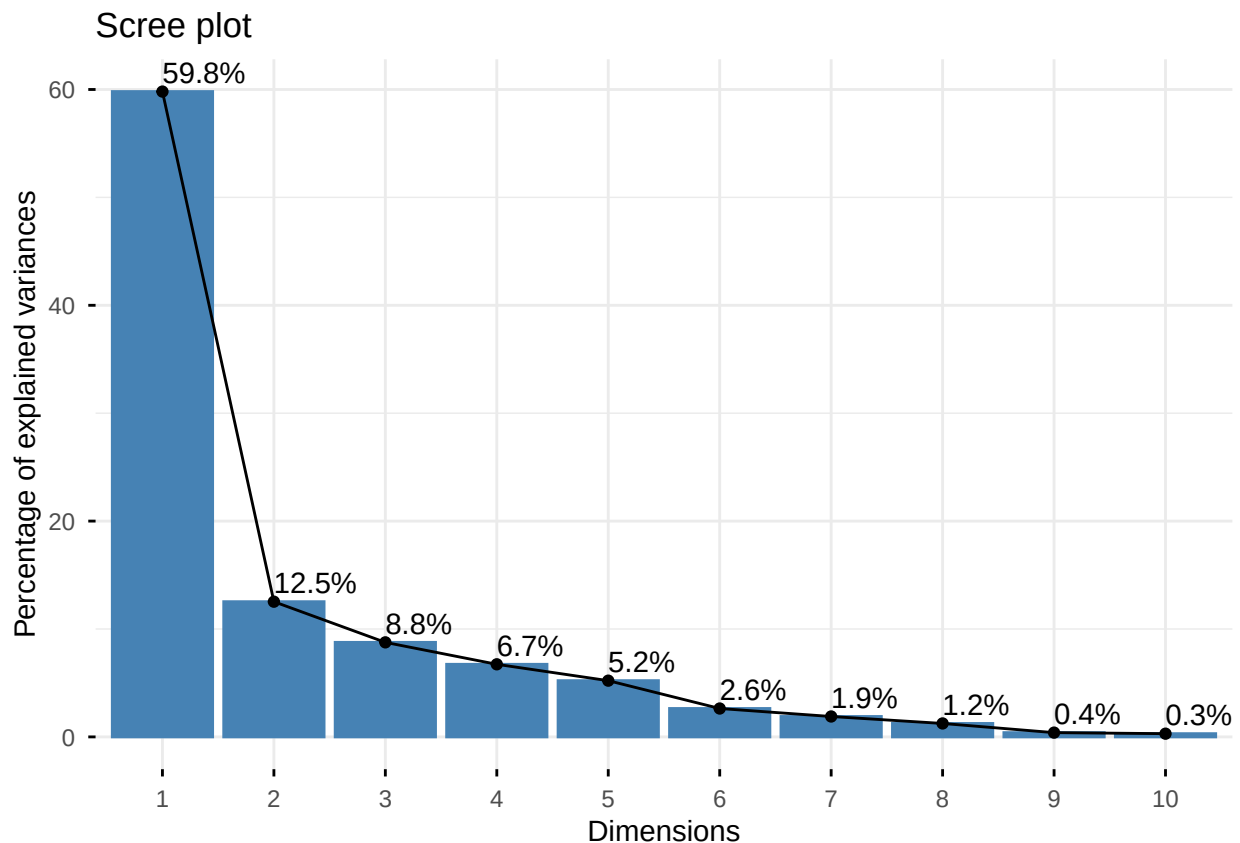
```
pc_scores <- as.data.frame(pca_res$x)
md <- mahalanobis(pc_scores[,1:2], colMeans(pc_scores[,1:2]), cov(pc_scores[,1:2]))
outliers <- states[md > qchisq(0.95, df=2)]
print(outliers)
```

```
## [1] "Jharkhand"      "Uttar Pradesh"
```

(c) Scree Plot

```
fviz_eig(pca_res, addlabels = TRUE)
```

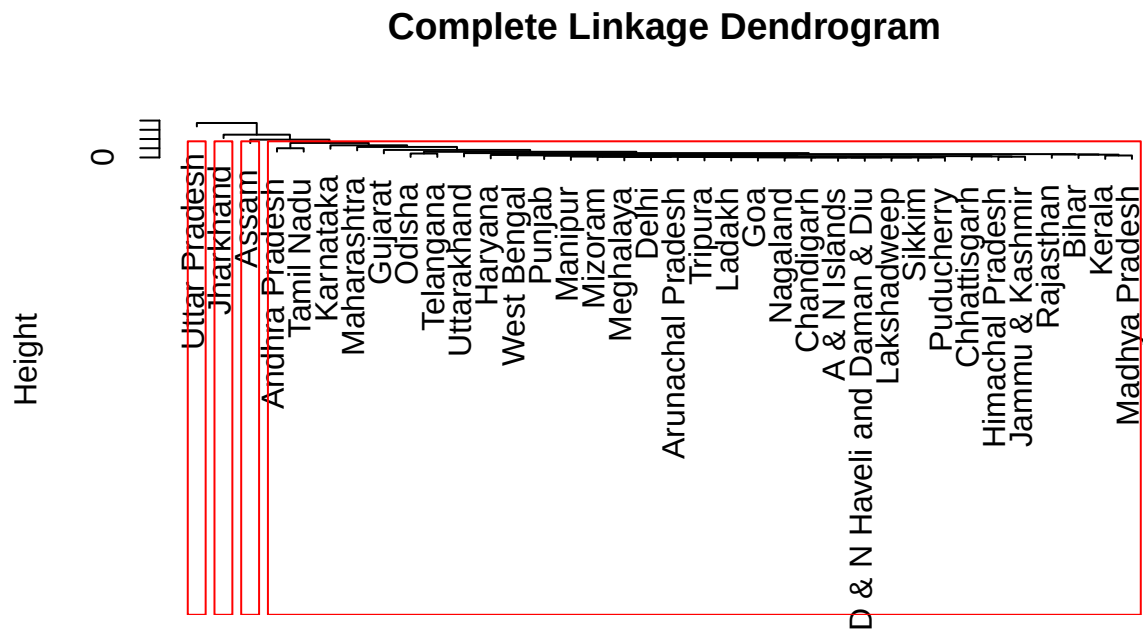
```
## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: 'width'.
```



Interpretation: The first two components explain a large portion of the variance. To achieve efficient dimensionality reduction, we should retain enough components to explain ~90% of the cumulative variance.

(d) Complete Linkage Hierarchical Cluster Dendrogram

```
dist_mat <- dist(data_scaled, method = 'euclidean')
hc_complete <- hclust(dist_mat, method = 'complete')
plot(hc_complete, main = "Complete Linkage Dendrogram", sub="", xlab="")
rect.hclust(hc_complete, k = 4, border = "red")
```



```
# List Cluster Members
hc_clusters <- cutree(hc_complete, k = 4)
for(i in 1:4) {
  cat(paste("\nCluster", i, ":", paste(states[hc_clusters == i], collapse = ", ")))
}
```

```
##
## Cluster 1 : Andhra Pradesh, Arunachal Pradesh, Bihar, Chhattisgarh, Goa, Gujarat, Haryana, Himachal Pradesh, Jammu & Kashmir, Jharkhand, Karnataka, Kerala, Lakshadweep, Madhya Pradesh, Maharashtra, Meghalaya, Mizoram, Nagaland, Odisha, Puducherry, Rajasthan, Sikkim, Tamil Nadu, Telangana, Tripura, Uttarakhand, Uttar Pradesh, West Bengal, and Yamantha Pradesh
## Cluster 2 : Assam
## Cluster 3 : Jharkhand
## Cluster 4 : Uttar Pradesh
```

(e) K-means Clustering (k=4)

```
set.seed(42)
km_res <- kmeans(data_scaled, centers = 4, nstart = 25)
fviz_cluster(km_res, data = data_scaled, repel = TRUE) +
  labs(title = "K-means Clustering (k=4)")
```

